UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	ICT450
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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QUESTION 1 (20 marks)

a) File system method is an improvement over manual system. However, there are many problems that associated with file system including data anomalies.

i) Briefly describe TWO (2) types of data anomalies.

(4 marks)

ii) Briefly describe TWO (2) applications of big data implemented in healthcare industry. Give an example for each application.

(4 marks)

b) Data redundancy is common issue in file system. Illustrate with description how data redundancy could occur.

(8 marks)

c) Based on the Entity Relationship Diagram in Figure 1.0, explain how referential integrity are enforced.

(4 marks)

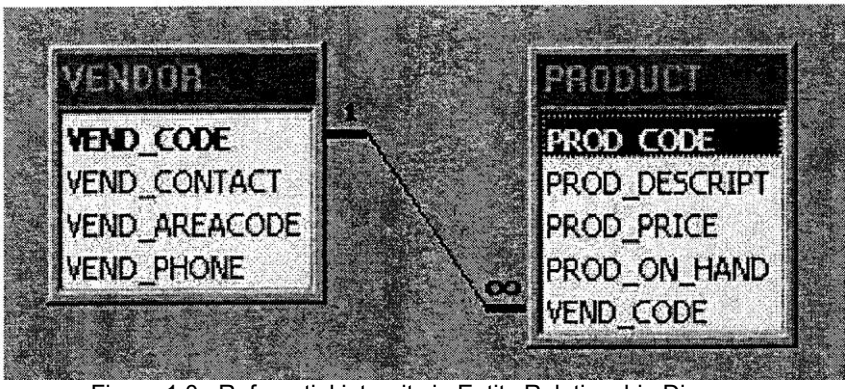


Figure 1.0 : Referential integrity in Entity Relationship Diagram

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QUESTION 2 **(20 marks)**

a) A good database design uses appropriate data models as its foundation. Data model also has been classified based on degrees of data abstractions by American National Standards Institute (ANSI) and Standards Planning and Requirements Committee (SPARC).

i) Give ONE (1) importance of data model to database system environment.

(2 marks)

ii) Explain with illustration each level of data abstraction.

(8 marks)

b) Database design may be based on two different database design strategies: centralized and decentralized. Discuss an appropriate situation for each the design strategy.

(4 marks)

c) State stages of Database Life Cycle (DBLC) in sequence.

(6 marks)

QUESTION 3 (20 marks)

The following questions will be based on data shown in Table 1.0, which was imported from an Excel spreadsheet. This represents a college course information that are offered. Each course is stored in a separate file. The management of data is complex and creates data redundancy. Therefore, to avoid from data redundancy the structure needs to be normalized.

Table 1.0 : Sample of college course information

Course Code	ICT450	Course Name	Database Design and Development	
Tutor ID	306612	Tutor Name	Professor Albus Dumbledore	
Student ID	Student Name	DOB	Gender	Last Attendance Date
2016123456	Luke Skywalker	11.10.96	F	31.01.18
2016534363	Kylo Ren	12.12.96	F	15.01.18
2016788924	Thor Loki	27.06.96	F	31.01.18
2016329874	Clint Barton	31.08.96	F	17.01.18
2016324793	General Hux	23.05.96	F	23.01.18
2016479625	Leia Organa	17.02.96	M	15.01.18
2016328796	Hermione Granger	15.04.96	M	31.01.18

a) State the normal form of the above Table 1.0. Explain your answer with reasons.

(3 marks)

b) Given that information, draw its dependency diagram and identify all transitive and partial dependencies.

(7 marks)

c) Normalize the table until third normal form (3NF). Show all your work.

(10 marks)

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QUESTION 4 **(20 marks)**

Consider the following relational database:

```
PUBLISHER (PublisherName, City, State, Country, PhoneNum)

AUTHOR (AuthorNum, AuthorName)

BOOK (ISBN, BookName, PublicationYear, AuthorNum, PublisherName)

CUSTOMER (CustNum, CustName, Street, City, State, Country, Email)

ORDER (OrderID, ISBN, CustNum, Date, Quantity)
```

a) Create table BOOK.

(4 marks)

b) Display the book number, book name and author name of all the books published by John Wiley & Sons Inc and Pustaka Berhad. View the results by book name in descending order.

(4 marks)

c) Find how many books that have been published in 2021 by Pustaka Trading located in Jasin, Melaka.

(4 marks)

d) Find the publisher which successfully published more than 50 books

(4 marks)

e) Find the book name which is total order is more than 3.

(4 marks)

QUESTION 5 (20 marks)

Chanteq Boutique is a local dress manufacturer. It sells its products to major distributors who distribute through their own stores in major cities and shopping malls. The primary product line is a casual dress for women.
Chanteq Boutique is in the process of designing a database system. The requirements needed are as following:

- Chanteq Boutique manufactures several products. Product id is a unique identifier. Product name is also need to be stored in this system.
- Each product has been assigned to a different stock-keeping unit (SKU). Each SKU belongs to a specific product. SKU number is a unique identifier of SKU. SKU description is also recorded in this table.
- Chanteq Boutique has created several unique colors for its dress. Each color is uniquely identified by a color id. A color description will describe the color options (e.g. red, blue, navy). Each SKU has only one color. Some colors have been assigned to SKU or may never be used.
- Each retailer will have a contract with Chanteq Boutique based on products it will distribute. A retailer will distribute one or more products, and a particular product will be distributed by more than one retailer. Each retailer has a unique id, name and phone number.
- Retailers have stores, which is the place where they sell the skirt and other products. Each retailer has one or more stores. To identify a store in the Chanteq Boutique database for shipping purposes, all customers have agreed to provide their store id and customer id as confirmation. They also need to write their address.

Based on the given information:

a) Construct an Entity Relationship Diagram (ERD) complete with all the required tables, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

b) Briefly explain TWO (2) examples of reports that can be generated from ERD in (a).

(4 marks)

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